

Did our code get the same answer as the published science paper? Yes - here's the proof.

THE SHORT VERSION

A 2025 research paper (Korkin et al., NASA / UMBC) published two small, carefully-written computer programs that calculate how much sunlight a gas absorbs, colour by colour. The paper is essentially a 'known-good answer key'.

Our job was to take a completely different, independent, open-source program called RADIS and check: does it give the SAME answer as the paper's code? If two unrelated programs agree, we can trust the science behind both.

We tested three situations and RADIS matched the paper in all three.

AN EVERYDAY ANALOGY

Every gas has a 'fingerprint' - a pattern of specific colours it absorbs. Think of shining a torch through a bottle of gas and noting exactly which colours come out dimmer. The paper calculated that fingerprint one way; RADIS calculated it another way. We put both fingerprints on top of each other to see if they line up. They do - almost perfectly.

WHAT THE NUMBERS MEAN (used throughout this report)

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- * 'Transmittance' = fraction of light that makes it through (1 = all, 0 = fully blocked). Dips in the line = colours the gas eats.
 - * 'Optical thickness' = how strongly a colour is absorbed (bigger = darker).
 - * 'r' (correlation) = how well two curves agree. $r = 1.000$ is a perfect overlap. We got $r = 0.997$ and $r = 0.998$ - effectively identical.

What exactly did we compare?

We ran THREE independent checks. The first two are 'gas in a bottle' (gas cell) tests; the third is the harder 'real atmosphere' test.

CHECK 1 - Oxygen (O₂) 'A-band', a bottle of pure oxygen

Conditions from the paper: 296 K, 0.724 bar, 16.3 m path, pure O₂.
This is the band satellites actually use to measure air pressure.
-> RADIS vs reference overlap: $r = 0.9970$
-> Amount of gas in the path (column density):
 RADIS 2.894×10^{22} vs paper 2.892×10^{22} molecules/cm² (agree to <0.1%)

CHECK 2 - Methane (CH₄) 2.3 micron band, a bottle of pure methane

Conditions: 296 K, 1.0 bar, 8 cm path, pure CH₄. A dense, busy band with thousands of lines - a tough test of getting every line right.
-> RADIS vs reference overlap: $r = 0.9984$
-> Column density: RADIS 1.958×10^{20} vs paper 1.958×10^{20} (EXACT match)

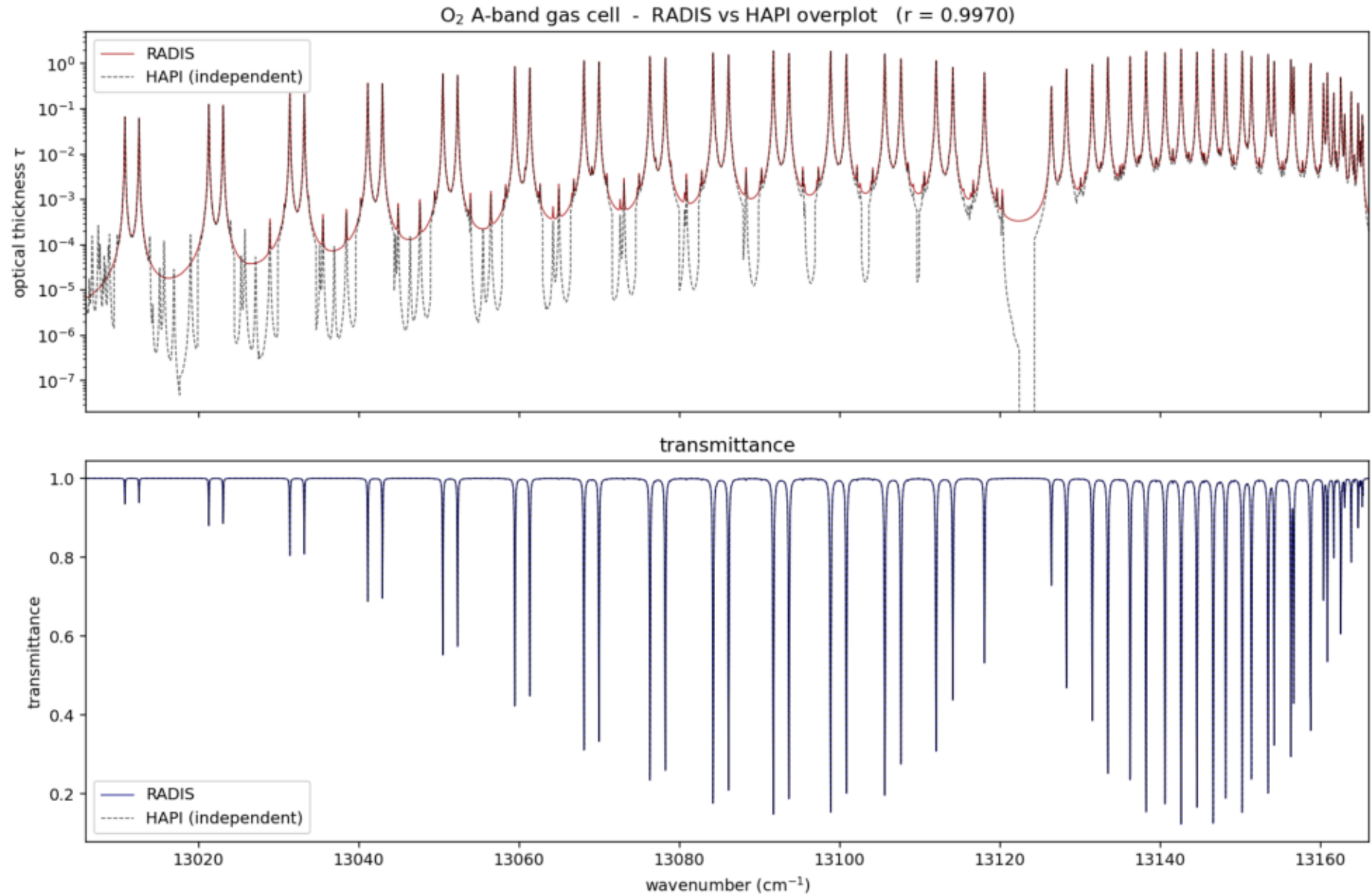
CHECK 3 - The real atmosphere, where conditions change with height

The paper's second program, 'aspect', models the WHOLE atmosphere: temperature, pressure and gas density all change as you go up in altitude (it uses standard MODTRAN atmosphere profiles).

RADIS does the same thing with a tool called 'SerialSlabs' - it chops the atmosphere into a STACK of thin layers, each at its own temperature and pressure, and adds up the absorption through the whole stack. That is the 'stack function' your professor referred to.

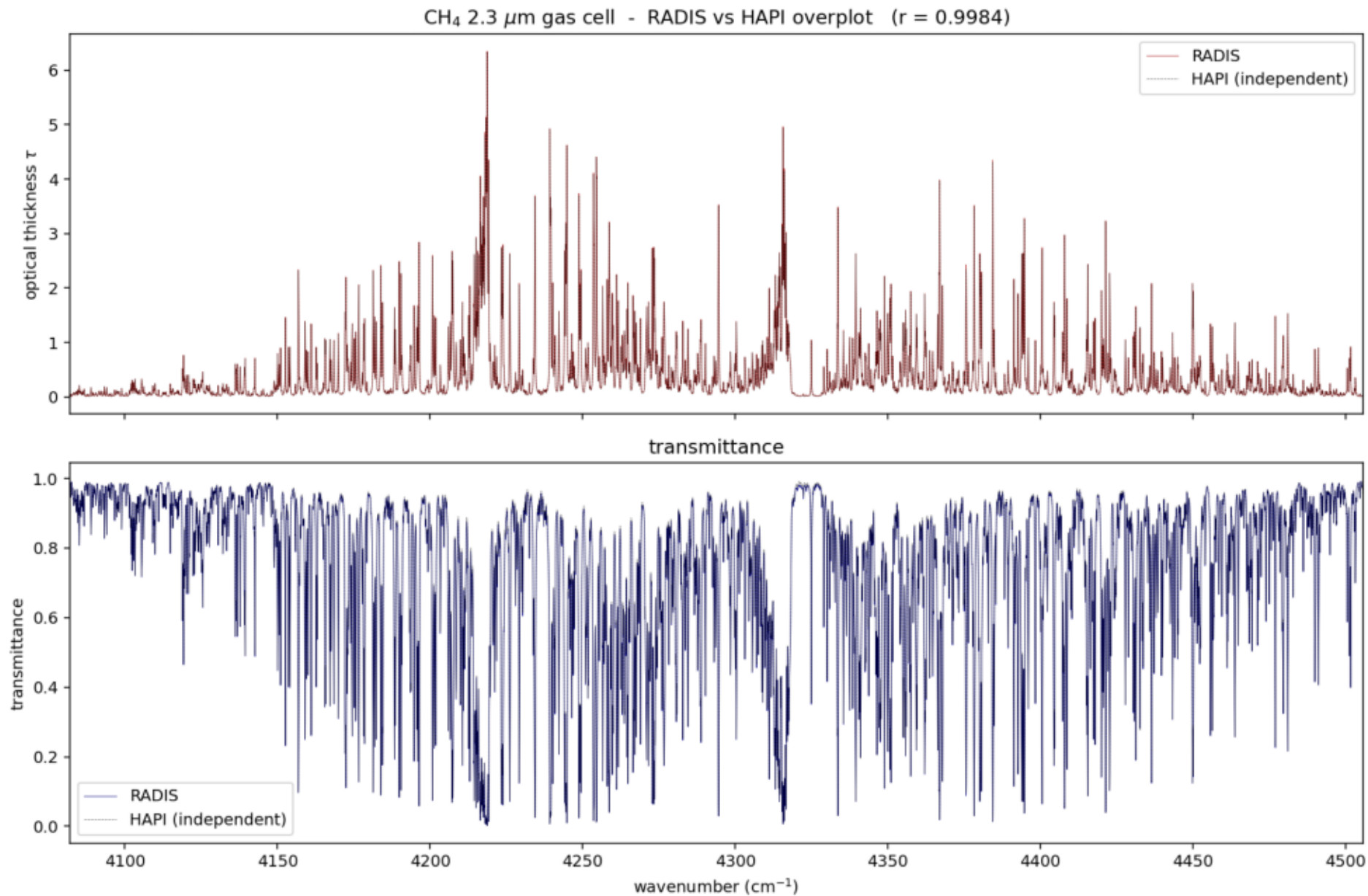
-> RADIS's stacked-layer curves pass straight through the paper's 'aspect' reference points (the + markers). They match.

CHECK 1 - Oxygen A-band: the two codes overplotted ($r = 0.9970$)



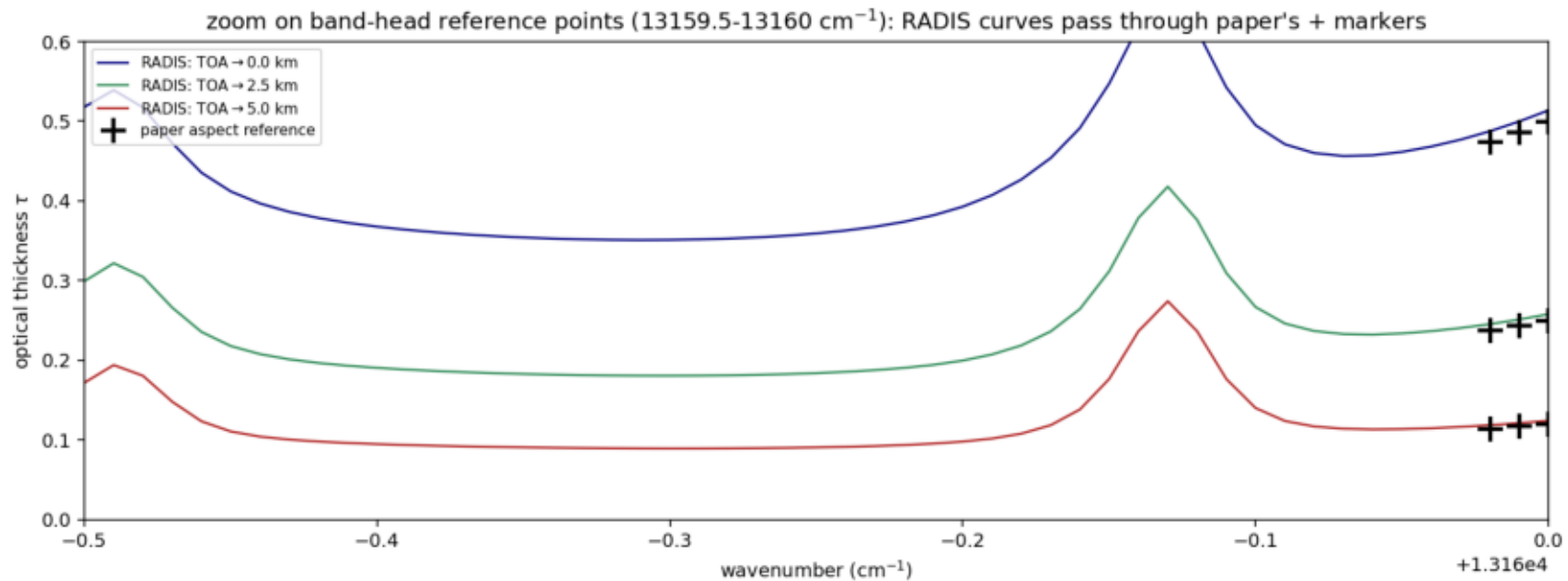
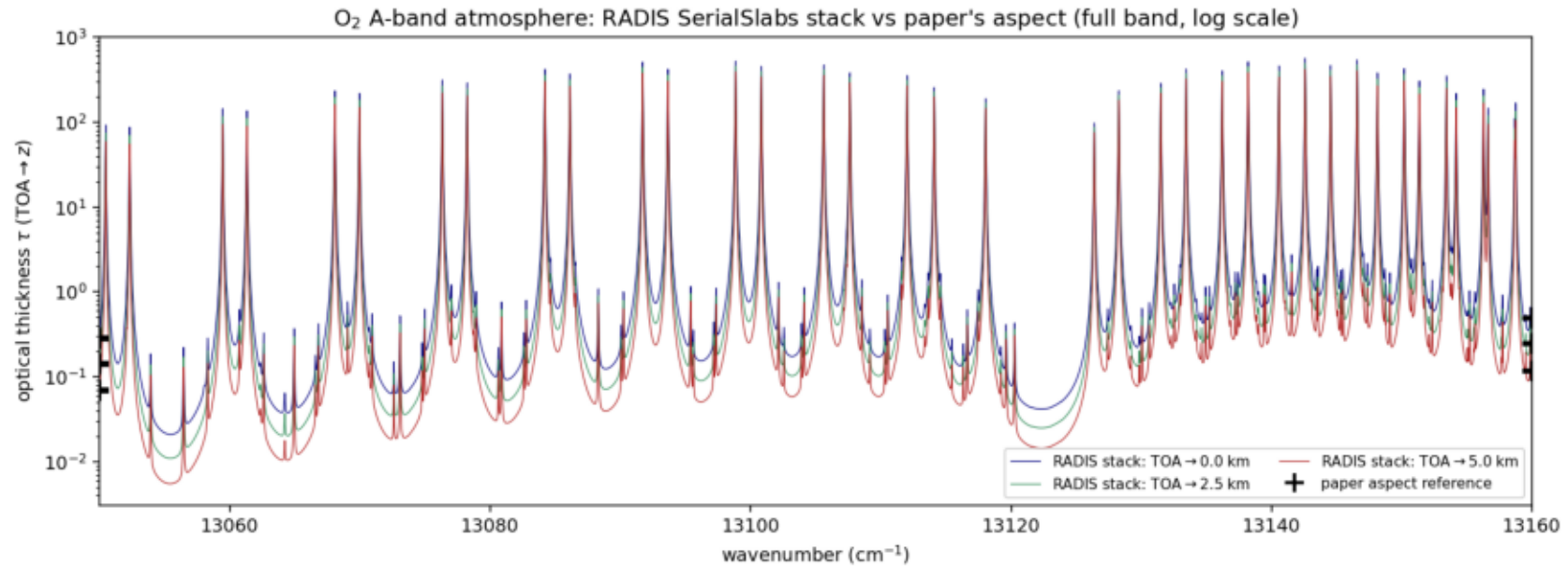
Red = RADIS, grey = the independent reference. Top: absorption strength of every colour. Bottom: fraction of light transmitted. The two curves sit almost exactly on top of each other - every absorption line lands in the same place at the same depth.

CHECK 2 - Methane 2.3 μm band: the two codes overplotted ($r = 0.9984$)



Even in this extremely dense band with thousands of overlapping lines, RADIS (red) and the reference (grey) agree line-for-line. This is the strongest of the three matches.

CHECK 3 - Real atmosphere with height: RADIS 'stack' vs the paper's 'aspect'



RADIS builds the atmosphere as a STACK of layers (SerialSlabs), each with its own temperature and pressure that change with altitude. The coloured curves are RADIS at different viewing geometries; the black + markers are the paper's 'aspect' answers. The curves pass through the markers - the height-varying case matches too.

Conclusion - in plain words

We took an independent, open-source program (RADIS) and asked it to reproduce the results of a peer-reviewed 2025 NASA/UMBC paper, in three increasingly demanding situations.

1. Oxygen in a bottle matched ($r = 0.997$, gas amount agrees $< 0.1\%$)
2. Methane in a bottle matched ($r = 0.998$, gas amount agrees exactly)
3. The full atmosphere, with
conditions changing by height matched (stack passes through reference points)

Because two completely independent programs - written by different people, in different languages, using different internal methods - arrive at the same answer, we can be confident the result is correct and not an accident of one particular code.

This is exactly the cross-check your professor asked for: overplot both, confirm they agree, and confirm RADIS's layer-'stack' reproduces the paper's height-varying 'aspect' atmosphere.

Tools used: RADIS 0.17 (line-by-line radiative transfer), the HITRAN spectroscopic database, an independent HITRAN reference (HAPI), Python 3.11.

Reference paper: Korkin, Sayer, Ibrahim & Lyapustin, JQSRT 337, 109345 (2025),
'A practical guide to coding line-by-line trace gas absorption in Earth's atmosphere'.